

```
In [31]: from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, classification_report, confusion_mat
import numpy as np
import pandas as pd
```

```
In [3]: df = pd.read_csv("age.csv")
print(df.head())
```

	Age	Income	Years_Employed	Satisfaction
0	22	250000	1	6
1	25	320000	2	7
2	28	450000	5	8
3	30	500000	6	7
4	35	750000	10	9

```
In [11]: features = ['Age', 'Income', 'Years_Employed']
X = df[features]
Y = df['Satisfaction']
```

```
In [15]: X_train, X_test, Y_train, Y_test = train_test_split(
    X, Y, test_size=0.2, random_state=42
)
```

```
In [21]: model = LogisticRegression(max_iter=1000)
model.fit(X_train, Y_train)
```

```
Out[21]: LogisticRegression
LogisticRegression(max_iter=1000)
```

```
In [25]: Y_pred = model.predict(X_test)
Y_pred
```

```
Out[25]: array([8, 6, 7, 8], dtype=int64)
```

```
In [33]: print("Accuracy:", accuracy_score(Y_test, Y_pred))

print("\nConfusion Matrix")
print(confusion_matrix(Y_test, Y_pred))

print("\nClassification Report")
print(classification_report(Y_test, Y_pred))
```

Accuracy: 0.0

Confusion Matrix

```
[[0 1 0 0]
 [0 0 0 1]
 [0 0 0 1]
 [0 0 1 0]]
```

Classification Report

	precision	recall	f1-score	support
5	0.00	0.00	0.00	1.0
6	0.00	0.00	0.00	1.0
7	0.00	0.00	0.00	1.0
8	0.00	0.00	0.00	1.0
accuracy			0.00	4.0
macro avg	0.00	0.00	0.00	4.0
weighted avg	0.00	0.00	0.00	4.0

```
C:\Users\mdars\anaconda3\Lib\site-packages\sklearn\metrics\_classification.py:150
9: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in label
s with no predicted samples. Use `zero_division` parameter to control this behavi
or.
```

```
_warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
```

```
C:\Users\mdars\anaconda3\Lib\site-packages\sklearn\metrics\_classification.py:150
9: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in label
s with no predicted samples. Use `zero_division` parameter to control this behavi
or.
```

```
_warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
```

```
C:\Users\mdars\anaconda3\Lib\site-packages\sklearn\metrics\_classification.py:150
9: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in label
s with no predicted samples. Use `zero_division` parameter to control this behavi
or.
```

```
_warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
```